ViruSafe-MPX for Reliability-Aware Differential Diagnosis of Mpox with Calibrated Uncertainty and External Validation

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***Abstract*—Mpox lesions can resemble chickenpox, measles, cowpox, hand-foot-and-mouth disease, and other dermatological conditions, making reliability as important as closed-set accuracy for image-based screening. This paper presents ViruSafe-MPX, a reliability-aware framework built on an EfficientNetV2-S visual encoder and a validation-selected multinomial reliability head operating on frozen deep features. The system combines probability calibration, selective prediction, class-conditional conformal sets, feature-space out-of-distribution detection, and duplicate-filtered external validation. On 755 MSLD images evaluated by leakage-resistant grouped five-fold out-of-fold testing, ViruSafe-MPX achieved 80.66 percent accuracy, 78.96 percent macro-F1, 95.50 percent macro-AUROC, and 74.65 percent Mpox recall. Selective accuracy reached 89.98 percent at approximately 70 percent coverage, while conformal prediction achieved 92.05 percent empirical coverage at a 90 percent target. On 653 duplicate-filtered external MSID images, restricted four-class accuracy was 76.42 percent and Mpox F1 was 83.54 percent. Feature-space OOD fusion improved unknown-case discrimination to a mean AUROC of 76.23 percent, although false-positive rates remained too high for autonomous rejection. The results support a bounded screening workflow in which uncertain or unfamiliar cases are referred rather than forced into a diagnosis.**

***Keywords*—Mpox, medical imaging, uncertainty calibration, conformal prediction, out-of-distribution detection, external validation**

I. INTRODUCTION

Mpox is a zoonotic viral disease whose cutaneous lesions can overlap visually with chickenpox, measles, cowpox, hand-foot-and-mouth disease (HFMD), and unrelated dermatological disorders. Laboratory confirmation remains central to diagnosis, but image-based artificial intelligence may still support screening and prioritization when confirmatory testing is delayed [1]–[3]. The challenge is that a conventional classifier is forced to choose among known labels even when the input is ambiguous or outside the training distribution.

Recent deep-learning studies have reported strong mpox image classification results using convolutional networks and vision transformers [4]–[7]. However, headline accuracy alone does not show whether confidence is calibrated, whether the system knows when to abstain, or whether performance persists on an independently sourced data set. These issues are especially important when web-sourced images can overlap between public data sets and when unfamiliar rashes may receive unjustifiably high confidence.

ViruSafe-MPX addresses this gap by treating classification as one component of a bounded reliability pipeline. Its contributions are: 1) leakage-resistant grouped out-of-fold evaluation on six MSLD classes; 2) a frozen-feature reliability head with validation-selected probability fusion; 3) calibration, selective referral, and class-conditional conformal prediction; 4) held-out-category OOD analysis using DermNet; 5) duplicate-filtered external validation on MSID; and 6) robustness and explanation-faithfulness checks. The aim is not to claim autonomous diagnosis, but to quantify when model output can be surfaced, when ambiguity should be communicated, and when referral is safer.

The paper therefore evaluates not only discrimination but also calibration error, classwise recall, risk-coverage behavior, conformal coverage, held-out OOD AUROC, duplicate-filtered external performance, corruption sensitivity, and Grad-CAM deletion faithfulness. This systems-level framing is deliberately more conservative than reporting a single favorable train-test split.

II. RELATED WORK

Ali et al. introduced MSLD v2.0 as a six-class mpox differential diagnosis benchmark and reported strong transfer-learning results [4]. Bala et al. developed MonkeyNet on the four-class MSID collection [5], while later studies examined transformer architectures and quantitative explainability for mpox lesion recognition [6], [7]. These works demonstrate useful visual signal but differ in data partitioning, augmentation, and duplicate control, so their headline metrics should not be treated as directly comparable leader-board scores.

The reliability modules used here draw on established methods beyond mpox classification. EfficientNetV2 provides an efficient visual representation [8]. Temperature scaling improves calibration without changing the predicted class [9]. Selective classification formalizes the accuracy-coverage trade-off [10], while conformal prediction yields set-valued uncertainty with empirically testable coverage [11]. OOD detection can be estimated from softmax confidence, feature geometry, energy scores, and deep nearest-neighbor distances [12]–[15]. ViruSafe-MPX integrates these mechanisms in a single evaluation protocol and tests them against external data rather than reporting them in isolation.

Explainability is another important distinction. Grad-CAM can highlight spatial regions associated with a prediction [18], but visually plausible heatmaps do not by themselves prove faithful reasoning. For that reason, this study restricts explanation examples to held-out images and supplements them with a deletion test that compares Grad-CAM-guided

masking with equally sized random masks. The reporting strategy also follows the transparency emphasis of TRIPOD+AI and the human-oversight perspective of DECIDE-AI [17], [19].

III. METHOD

A. Data and Evaluation Protocol

The development set was MSLD v2.0 with 755 original images from six classes: Mpox, chickenpox, measles, cowpox, HFMD, and healthy skin [4]. The public release used in this study yielded 524 recoverable patient groups after filename and duplicate auditing. Recoverable groups were kept intact during five-fold partitioning; each fold contained 483 training, 121 validation, and 151 test images. Every image contributed exactly once to pooled out-of-fold evaluation.

DermNet [20] was used only as an OOD source. Categories were divided into 8 method-selection categories and 14 disjoint held-out test categories. External validation used MSID version 6 [5], [16]. Exact SHA-256 and perceptual-hash filtering against MSLD removed exact and near-duplicate candidates before evaluation, leaving 653 usable external images from 770 discovered files. Table I summarizes the role of each data source.

TABLE I. DATA SOURCES AND EXPERIMENTAL ROLES

Source	Size / classes	Role
MSLD v2.0	755 / 6	Train, validation, grouped OOF test
DermNet	22 OOD categories	8 select + 14 held-out test
MSID v6	653 retained / 4	External validation only

Fig. 1 combines a clinical appearance montage of the six study categories with representative held-out Grad-CAM examples. The montage provides visual context for the differential task, while the attribution examples show a correct prediction, an uncertain case, and an error. The images are included for interpretability and not as a substitute for clinical examination.

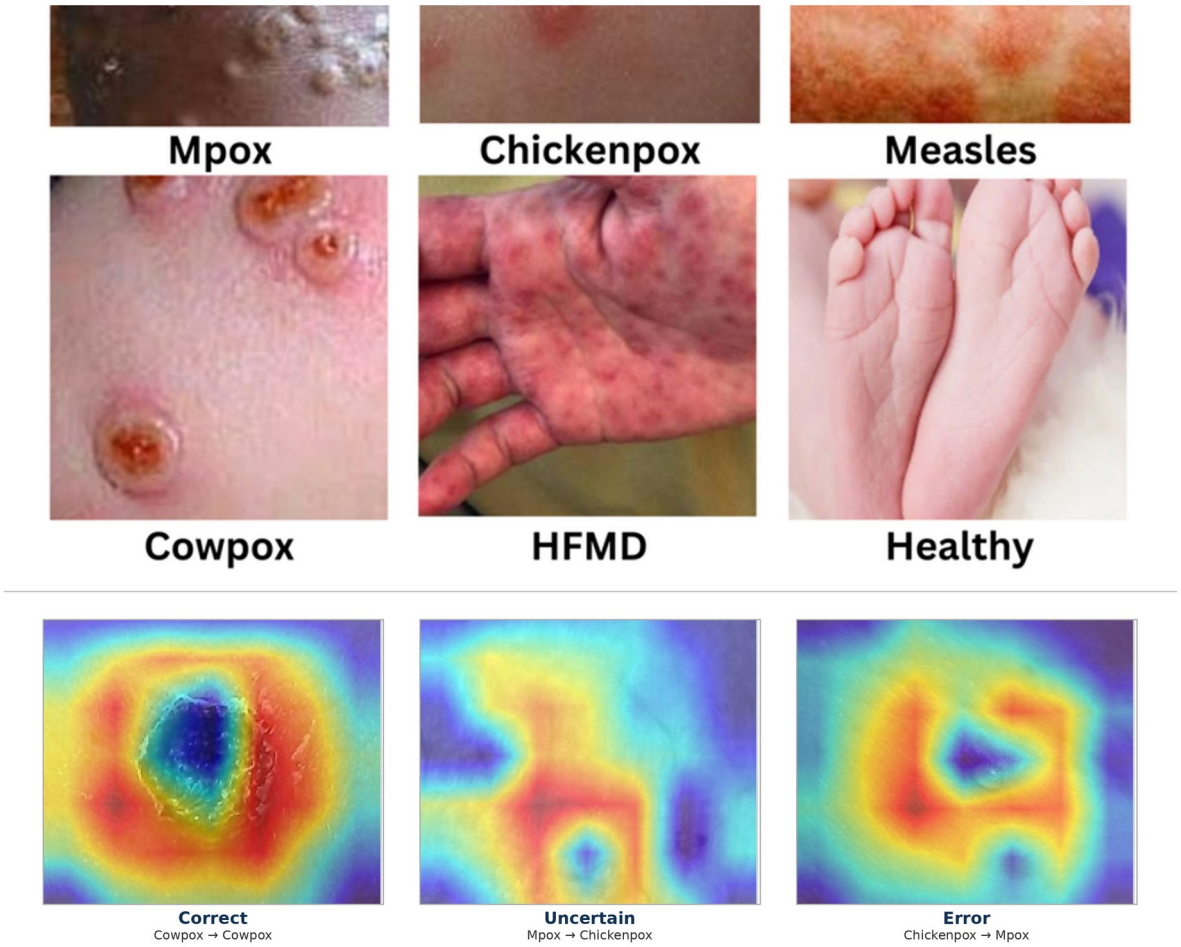


Fig. 1. Clinical appearance montage (top) and representative held-out MSLD Grad-CAM examples (bottom), including correct, uncertain, and error cases. Grad-CAM overlays were generated in this study; source photographs remain subject to the original image rights.

B. Backbone and Reliability Head

The baseline used ImageNet-pretrained EfficientNetV2-S [8] with 224 by 224 pixel inputs. Training used class-weighted cross-entropy with 0.05 label smoothing, AdamW with a learning rate of 1.5 times 10 to the minus 4, weight decay of 10 to the minus 4, mixed precision, gradient clipping, cosine annealing, and early stopping on validation macro-F1. Moderate crop, flip, rotation, color-jitter, and perspective augmentation were applied only to development data.

For v1.1, the CNN was frozen. Its 1,280-dimensional features were standardized and passed to a multinomial logistic-regression reliability head. Regularization and class weighting were chosen by grouped inner cross-validation using training embeddings only. CNN and reliability-head probabilities were then blended with a weight selected on the outer validation partition from 0, 0.25, 0.50, 0.75, and 1.0. The enhanced predictor was activated only when validation macro-F1 improved by at least 0.002.

Fig. 2 summarizes the inference pipeline. The resulting probabilities are calibrated, then exposed to three reliability mechanisms: class-conditional conformal sets, confidence-based selective referral, and a feature-space OOD warning.

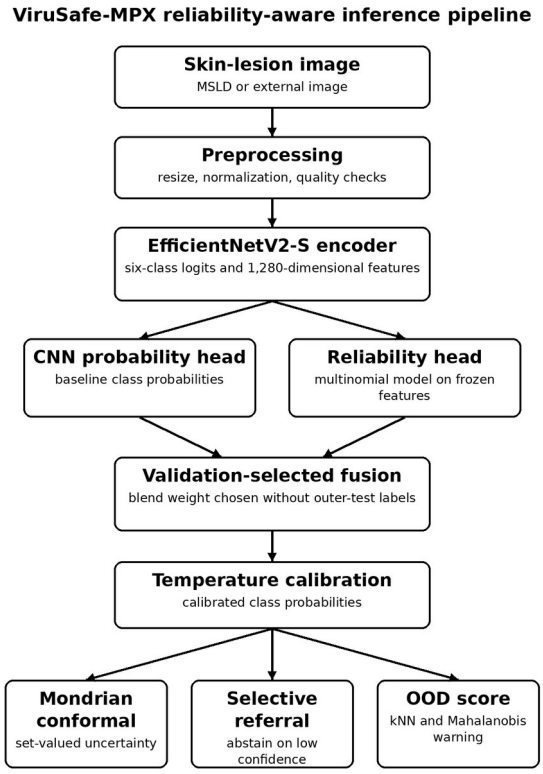


Fig. 2. ViruSafe-MPX reliability-aware inference pipeline. Model selection and probability fusion use development and validation data only; outer-test labels are not used for tuning.

C. Calibration, Selective Prediction, and Conformal Sets

A scalar temperature was fitted on each outer validation partition by minimizing multiclass negative log-likelihood [9]. Selective prediction then ranked test examples by maximum calibrated probability and evaluated retained coverage at 70, 80, 90, and 95 percent [10]. Rejected examples represent a referral set rather than an automatic negative diagnosis.

Class-conditional Mondrian conformal prediction used validation nonconformity scores separately for each disease class [11]. At a nominal 90 percent target, each candidate class was included in the prediction set when its score was below the class-specific finite-sample threshold. Empty sets were replaced by the top-1 class and recorded as fallback events.

D. OOD Detection and External Validation

Candidate OOD scores included maximum softmax probability, entropy, margin, energy, cosine prototype distance, PCA-based Mahalanobis distance, relative Mahalanobis distance, cosine k-nearest-neighbor distance, and pre-specified score fusions [12]–[15]. The method was selected only on the 8 OOD-validation categories; the final score was frozen before evaluation on the 14 held-out categories. The selected method was an equal-weight fusion of cosine k-nearest-neighbor distance and PCA-64 Mahalanobis distance.

For MSID, five fold-specific models were averaged as an ensemble. Two evaluations were retained: strict six-class accuracy, where cowpox and HFMD remained possible outputs, and restricted four-class accuracy over the labels shared with MSID. This preserves the external data set as a genuine domain-shift test rather than a source of model tuning.

E. Robustness, Explainability, and Statistical Analysis

Controlled corruption tests were performed on the fold-1 backbone at three severities for brightness shift, blur, JPEG compression, occlusion, and Gaussian noise. These tests are diagnostic rather than the primary pooled v1.1 endpoint because they predate the frozen-feature reliability extension. Explainability used Grad-CAM on 12 held-out fold-1 test images sampled from high-confidence correct, uncertain, and error groups [18]. Faithfulness was probed by masking the top 10, 20, and 30 percent of Grad-CAM-ranked pixels and comparing the target-confidence drop with random masking of the same size.

Paired model improvement was tested with a patient/group-cluster bootstrap on pooled out-of-fold predictions and McNemar's exact test. This pairing is important because baseline and v1.1 predictions are available on the same 755 outer-test examples; gains can therefore be evaluated as changes on identical held-out cases rather than as differences between unrelated folds.

IV. RESULTS

A. Internal Out-of-Fold Performance

ViruSafe-MPX v1.1 improved pooled accuracy from 73.51 to 80.66 percent, balanced accuracy from 76.83 to 79.45 percent, macro-F1 from 72.92 to 78.96 percent, and macro-AUROC from 94.25 to 95.50 percent. Mpox recall increased from 55.63 to 74.65 percent. Calibration also improved, with expected calibration error decreasing from 4.44 to 3.65 percent and Brier score from 0.380 to 0.298. Fig. 3 shows the pooled performance and six-class confusion pattern, while Table II reports the primary pooled metrics.

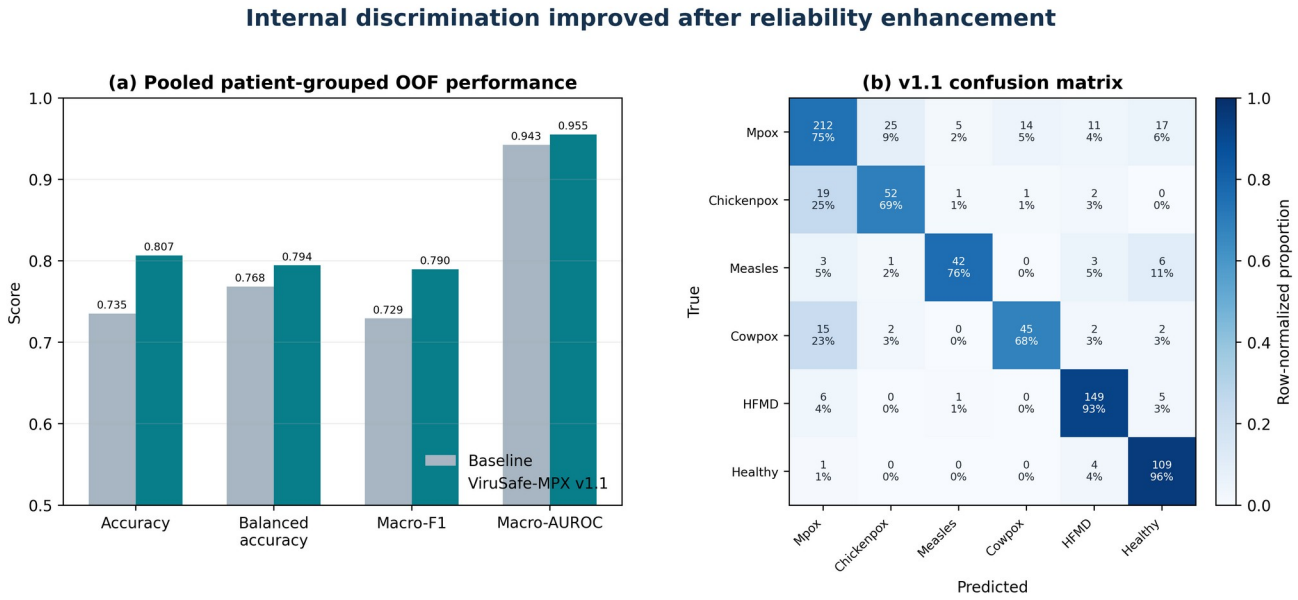


Fig. 3. Pooled out-of-fold performance and row-normalized confusion matrix for ViruSafe-MPX v1.1. Each MSLD image contributes once as a held-out outer-fold prediction.

TABLE II. PRIMARY INTERNAL AND RELIABILITY RESULTS

Metric	Baseline	v1.1
Accuracy	73.51%	80.66%
Macro-F1	72.92%	78.96%
Macro-AUROC	94.25%	95.50%
Mpox recall	55.63%	74.65%
ECE	4.44%	3.65%
Brier score	0.380	0.298

The paired improvement was statistically supported. Patient/group-cluster bootstrap analysis estimated a mean accuracy gain of 7.12 percentage points with a 95 percent confidence interval from 4.33 to 9.74 points, and a macro-F1 gain of 6.05 points with a 95 percent confidence interval from 3.30 to 8.68 points. McNemar testing found 68 baseline errors corrected by v1.1 versus 14 new errors, giving an exact p value of 1.13 times 10 to the minus 9. Table III reports class-wise precision, recall, and F1.

TABLE III. CLASS-WISE POOLED PERFORMANCE OF VIRUSAFE-MPX V1.1

Class	Precision	Recall	F1
Mpox	82.81%	74.65%	78.52%
Chickenpox	65.00%	69.33%	67.10%
Measles	85.71%	76.36%	80.77%
Cowpox	75.00%	68.18%	71.43%
HFMD	87.13%	92.55%	89.76%
Healthy	78.42%	95.61%	86.17%

HFMD and healthy skin were the strongest classes, whereas chickenpox remained the most difficult infectious differential. Mpox F1 increased to 78.52 percent with 82.81 percent precision and 74.65 percent recall. The remaining confusions motivate the set-valued and referral mechanisms described below.

B. Reliability and Selective Referral

Reliability gains were most visible when abstention was permitted. At approximately 70 percent retained coverage, selective accuracy reached 89.98 percent and selective macro-F1 88.97 percent. At 90 percent coverage, accuracy remained 84.71 percent. The corresponding baseline accuracies were 85.63 and 78.24 percent. At the same time, class-conditional conformal prediction achieved 92.05 percent pooled coverage at a nominal 90 percent target, with a mean set size of 1.93 labels and Mpox coverage of 92.25 percent. Fig. 4 illustrates the accuracy-coverage trade-off and calibration improvement.

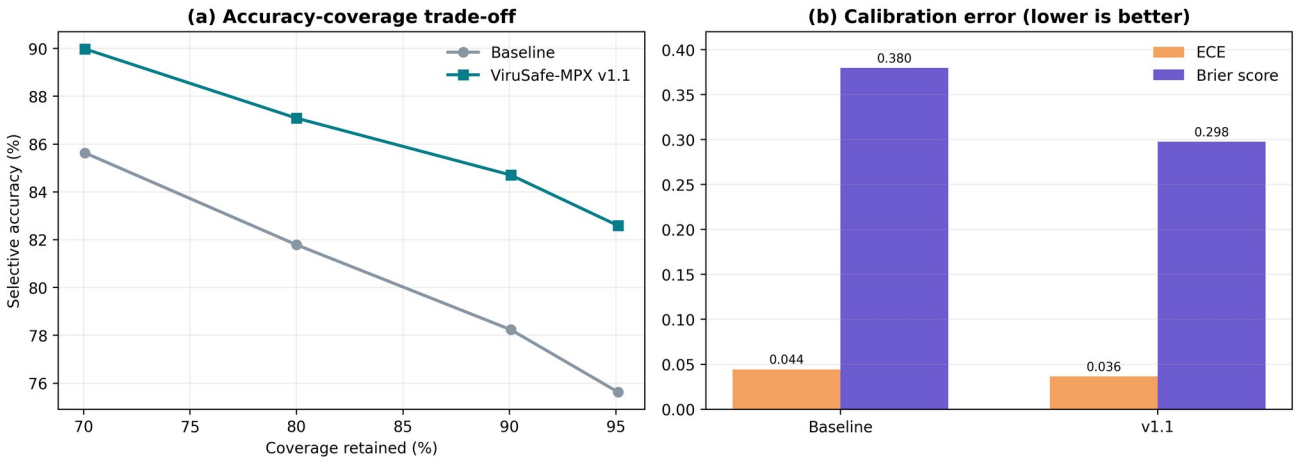


Fig. 4. Reliability-aware selective prediction and calibration. Withholding lower-confidence cases improves the accuracy of retained predictions, while the v1.1 probability layer reduces calibration error.

Conformal behavior was also class dependent. At a nominal 90 percent target, pooled coverage was 92.05 percent with a mean prediction-set size of 1.93 labels. Mpox coverage was 92.25 percent; chickenpox, measles, and HFMD also exceeded 90 percent, while healthy and cowpox were slightly lower at 89.47 and 87.88 percent. Fig. 5(a) reports this profile alongside the held-out-category OOD AUROC across folds.

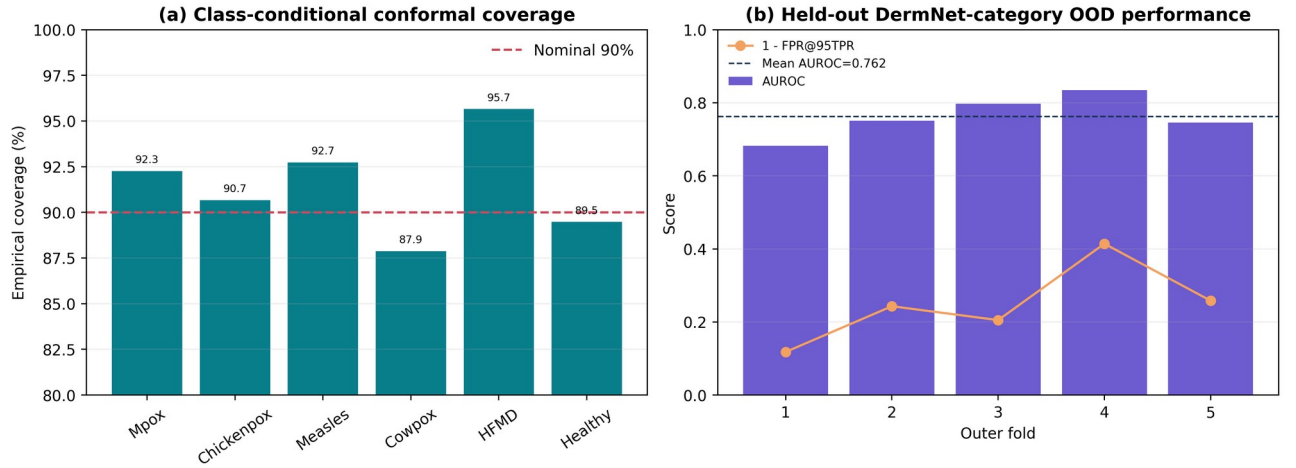


Fig. 5. Uncertainty sets and open-set evaluation: class-conditional conformal coverage relative to the 90 percent target and held-out DermNet-category OOD AUROC across five outer folds.

C. *Open-Set, Robustness, and External Validation*

Across the five folds, the selected feature-space OOD fusion achieved a mean held-out-category AUROC of 76.23 percent with a standard deviation of 5.76 percent. At thresholds designed to accept approximately 95 percent of in-distribution validation examples, mean OOD rejection was only 27.12 percent and mean FPR at 95 percent TPR was 75.23 percent. Thus, OOD scoring is useful as a warning signal but is not reliable enough to autonomously reject unknown dermatological conditions.

Image-quality stress testing revealed a complementary failure mode. Mild brightness changes, blur, JPEG compression, and occlusion preserved macro-F1 near the uncorrupted operating range, but severe corruption caused larger losses. Severe blur and JPEG compression reduced macro-F1 to 57.61 and 57.13 percent, respectively, and severe Gaussian noise reduced it to 36.69 percent. Fig. 6 summarizes these trends and motivates an image-quality gate before inference.

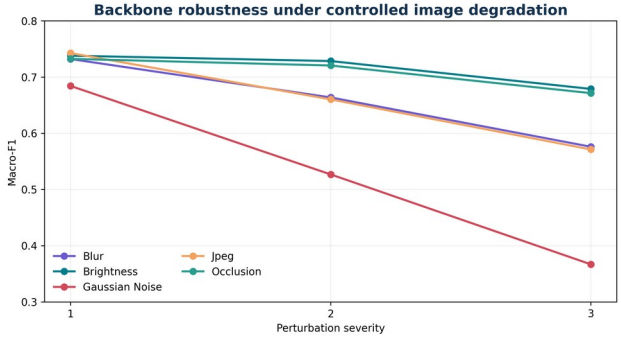


Fig. 6. Backbone robustness under controlled image degradation at three perturbation severities.

External MSID evaluation was performed after duplicate filtering. The v1.1 ensemble achieved 71.36 percent strict six-class accuracy and 76.42 percent restricted four-class accuracy, with a four-class macro-F1 of 71.62 percent. External Mpox precision, recall, and F1 were 95.38, 74.32, and 83.54 percent, respectively. Chickenpox remained the main external weakness, with an F1 of 54.76 percent. Fig. 7 shows the external confusion pattern and class-wise F1 after

duplicate removal; Table IV summarizes open-set and external-validation outcomes.

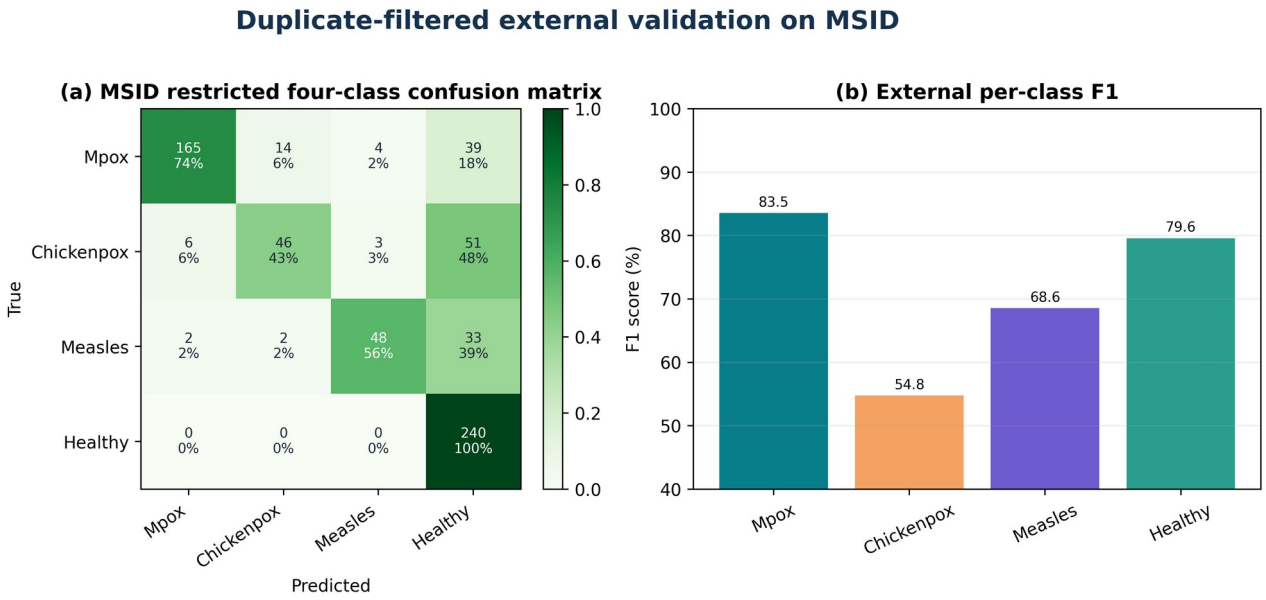


Fig. 7. Duplicate-filtered external validation on MSID: restricted four-class confusion matrix and class-wise F1 for the v1.1 ensemble.

TABLE IV. OPEN-SET AND EXTERNAL VALIDATION SUMMARY

Evaluation	Result	Interpretation
OOD AUROC	76.23% ± 5.76	Moderate unseen-category separation
OOD FPR@95TPR	75.23%	Too high for autonomous rejection
MSID strict 6-class	71.36% acc.	Full output label space
MSID restricted 4-class	76.42% acc.	71.62% macro-F1
External Mpox	83.54% F1	95.38% precision; 74.32% recall

D. Explainability and Ablation

Held-out Grad-CAM examples were deliberately drawn from correct, uncertain, and erroneous predictions. This avoids presenting explanation only on favorable cases. Across 12 test examples, masking the top 30 percent of Grad-CAM-ranked pixels reduced target-class confidence by 0.249 on average, compared with 0.223 for random masking. The difference supports some localization faithfulness, but its modest size means that attribution should remain an audit aid rather than causal proof. Fig. 8 summarizes the held-out explanation examples and deletion-faithfulness curves.

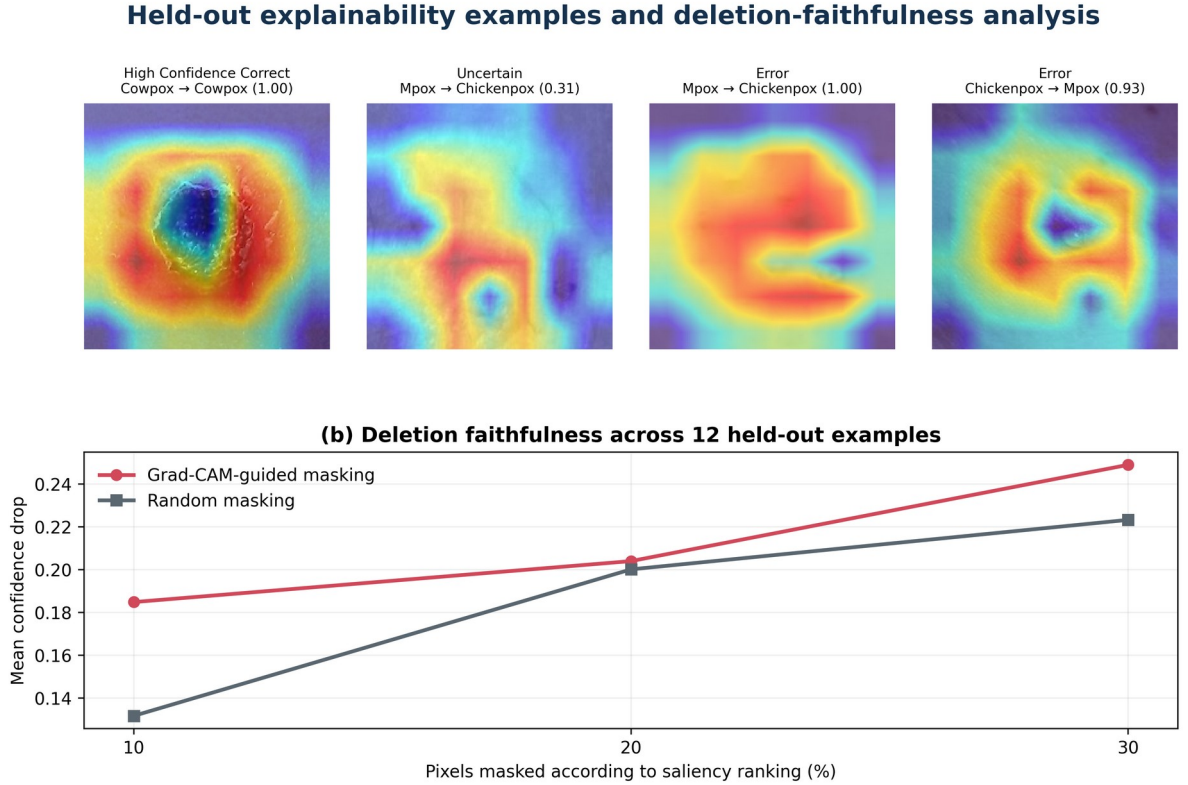


Fig. 8. Held-out Grad-CAM examples and deletion-faithfulness analysis. Saliency-guided masking causes a larger mean confidence drop than equally sized random masks at the highest tested masking level.

Ablation showed that the frozen-feature reliability head and validation-selected fusion produced the main classification gain: accuracy increased from 73.51 to 80.66 percent, Mpox recall from 55.63 to 74.65 percent, and selective accuracy at 90 percent coverage from 78.24 to 84.71 percent. Conformal prediction and the OOD module then add set-valued uncertainty and open-set warning information rather than artificially increasing closed-set accuracy.

V. DISCUSSION

The main contribution of ViruSafe-MPX is not a single peak accuracy value. The more important result is that a standard visual backbone becomes more useful when uncertainty and failure handling are made explicit. The frozen-feature reliability head improved both discrimination and calibration without another CNN training cycle, while selective prediction showed a clear accuracy-coverage benefit. Conformal sets further allowed the system to express ambiguity, for example by retaining more than one plausible disease label instead of forcing a single prediction.

Published mpox classifiers report higher raw closed-set accuracy [4]–[7], but their class spaces, partitioning, augmentation, and external-test definitions differ. ViruSafe-MPX instead prioritizes grouped out-of-fold evaluation, calibrated uncertainty, abstention, held-out-category OOD testing, and duplicate-filtered external validation; its internal accuracy is therefore a conservative systems-level estimate rather than a direct leaderboard claim.

External validation is equally important. Public skin-image data sets can share web-sourced photographs, so an external test may look stronger than it is if duplicates are not removed. The duplicate-filtered MSID result is therefore a stronger generalization test than a random internal split. At the same time, the lower external chickenpox performance shows that domain shift remains clinically relevant.

The OOD analysis also exposes a limitation rather than hiding it. Feature-space fusion materially improved unknown-case separation over confidence-only baselines, but the operating false-positive rate remained high. ViruSafe-MPX should therefore treat OOD as one input to a referral policy, not as a definitive gate. Other limitations include the modest size of MSLD, incomplete recoverability of patient identity metadata, internet-sourced images, lack of demographic subgroup analysis, and absence of prospective clinical acquisition. Grad-CAM and deletion testing provided supportive explainability evidence, but attribution maps do not establish causal reasoning or clinical validity.

These findings suggest a deployment pathway in which an image-quality check precedes inference, calibrated probabilities and conformal sets are shown to the operator, and low-confidence or OOD-like cases are routed to expert assessment and PCR confirmation. This bounded workflow is consistent with recent guidance emphasizing transparent evaluation and human oversight for clinical prediction systems [17], [19].

VI. CONCLUSION

ViruSafe-MPX combines differential lesion classification with calibration, selective referral, conformal prediction, feature-space OOD analysis, and independent external validation. The framework achieved 80.66 percent pooled six-class accuracy, 78.96 percent macro-F1, and 95.50 percent macro-AUROC, while improving Mpox recall to 74.65 percent. At 70 percent coverage, selective accuracy reached 89.98 percent; on duplicate-filtered external MSID data, four-class accuracy was 76.42 percent and Mpox F1 was 83.54 percent. These results support reliability-aware screening rather than autonomous diagnosis. Future work should focus on prospectively acquired multi-center images, richer clinical metadata, fairness analysis, and pre-specified human-in-the-loop operating thresholds.

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